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Effects of the flipped classroom model on student performance and interaction with a peer-coach strategy

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ABSTRACT

A peer-coaching approach was adopted in the out-of-class stage of the flipped classroom. Out of the classroom, learning tutors help students engage in learning and improve performance. Eighty-seven participants from two classes were involved in this study. One class of 44 students was the experimental group with the peer-coaching approach in a flipped classroom, and the other class of 43 students served as the control group learning by the conventional video-based flipped classroom mode. The experimental results revealed that peer coaching improved student engagement and interaction out of class. In the experimental group, students interacted not only with tutors and group peers but also with students from other groups. The experimental results also show that the proposed approach could improve students' learning achievements.

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KEYWORDS

Flipped classroom; flipped learning; peer coach; collaborative learning

Introduction

The flipped classroom model is initially within secondary maths and science classrooms (e.g. Bergmann and Sams 2012a), but it is widely implemented in higher education (Tsai et al. 2019; Winter 2018; Zainuddin and Halili 2016) and has become more prominent in K-12 education (Carhill-Poza 2019; Cui and Yu 2019). In a flipped classroom environment, students can access the learning content and instruction online and on their own time and can control the pace at which they move through the online elements before coming to class (Bergmann and Sams 2012a; Galway et al. 2014; Kazanidis et al. 2019). During class, students apply the learning content to solve problems or participate in discussions with the instructor's guidance. In the classroom, researchers argued that students could engage in active learning and deepen their understandings via learning tasks such as debates, case studies and collaborative work (Gilboy, Heinerichs, and Pazzaglia 2015; Holmes et al. 2015). These learning tasks often require students to properly process information from different sources and then critically assimilate information for knowledge construction (Siu 2014).

Hao's (2016a) study showed that most students had particular preferences for the flipped classroom. Some researchers suggested that the flipped classroom pedagogy was able to promote learning outcomes (Sergis, Sampson, and Pelliccione 2018), help students develop creativity (Al-Zahrani 2015), and lead to greater classroom engagement and better professional skills development (Elmaadaway 2018). However, Boevé et al. (2016) argued that students' study behaviours in the flipped classroom did not appear to be very different from those of students in a regular course, and study behaviours did not appear strongly related to student performance in the flipped classroom. In theory, when students are encouraged to engage and complete learning activities out of class, the positive effects of the flipped learning may be realised (Huang and Hew 2018). Hao (2016b) stated that students need to access the flipped learning material and engage with learning activities if they want to achieve better learning outcomes. Moreover, students may learn how to manage their time working with the online course and develop self-study and autonomous learning skills making them active participants of the educational process.

For teachers, however, there are some challenges to managing students outside the classroom. Teachers should expend more time and energy to tutor students when they experience problems with understanding the learning materials out of class. Many students fail to complete learning tasks out of class because they lack the competence in learning control, self-direction, and self-regulation capacity. Ultimately, they may not be ready for the learning activities in class (Hao 2016b; Lai and Hwang 2016).

To help students gain a deeper understanding of what they are learning, and complete the learning tasks prior to class, a peer-coach strategy was applied in the flipped classroom in this study. Peer coaching is a process in which a coach regularly observes a peer, and the coach provides support, assistance, and feedback to the peer (Hagen, Bialek, and Peterson 2017). For peer coaching, the coach and his/her peer work together to share ideas, support each other, teach one another, reflect on current practices, build new skills or solve problems (Hohensee and Lewis 2019; Joyce and Showers 1980). Peer coaching could improve the teaching practice (Sider 2019), refine learners' professional skills (Bowman and McCormick 2000) and enhance learners' engagement in learning (Olpak, Baltaci, and Arican 2018). Thus, in the current study, we aimed to assess whether the peer-coaching strategy could enhance students' interactions and improve students' academic performances in the flipped classroom.

Literature review

Flipped classrooms

In a flipped classroom, students gain basic knowledge out of the class and engage in more learning activities in the class (Hwang and Lai 2017). In this learning model, students take lectures via instructional videos or other learning materials, at their own pace, prior to the class, and complete their homework or do exercises in the class with help from the teacher or peers. Instructors encourage students to engage in active learning and knowledge building through whole-class brainstorming, group-based hands-on assignments and peer reviews, feedback exchange and remedial assistance (Bergmann and Sams 2012b; Davies, Dean, and Ball 2013).

Several studies have been conducted on flipped classrooms in K-12 education and higher education environments to investigate the aspects of the strategy's effectiveness (Alsaleh 2019; Findlay-Thompson and Mombourquette 2014; Lai and Hwang 2016; Lew 2016; Wang 2017; Zainuddin and Perera 2019). Most of these have shown promising results. For K-12 students, the flipped classroom model significantly benefitted their learning achievement, motivation and self-efficacy (Bhagat, Chang, and Chang 2016; Kostaris et al. 2017; Lai and Hwang 2016). Flipped classrooms could increase academic subject satisfaction and teaching effectiveness (Tse, Choi, and Tang 2019). Chang and Hwang (2018) conducted experimental research on fifth-grade students and found out that the augmented reality-based flipped learning guiding approach not only benefitted the students in terms of promoting their project performances but also improved their learning motivations, critical thinking tendencies, and group self-efficacy. Sergis, Sampson, and Pelliccione (2018) took experimental research on junior high school students (equivalent to the 8th grade; students approximately 14 years old) and high school students (equivalent to the 10th grade; students approximately 16 years old) and found that flipped classrooms not only increased students' cognitive learning outcomes, but also increased their internal dispositions.

For higher education, in flipped classrooms, students were found to be more proficient and achieve significantly better learning achievements when compared to students in conventional classrooms (Alsaleh 2019; Mason, Shuman, and Cook 2013; Prashar 2015; Wang 2017). Butt (2014) conducted surveys on college students participating in flipped classrooms and found that students held a positive attitude towards flipped classrooms, and more than 75% of the students believed that flipped classrooms were conducive to their learning. Adnan (2017) studied 70 senior students (thirty-one students in the experimental group with an average age of 22.8; thirty-nine students in control group with an average age of 23.4; the average age of all the students was 23.2) and found that the scores of the flipped group were significantly higher than those of the nonflipped group, and the students had positive cognition and satisfactory learning experiences with flipped learning. According to research on a flipped classroom in a college English course, the flipped classroom could improve students' English performances, strengthen the communications and interactions between students, and improve students' oral abilities (Abdullah, Hussin, and Ismail 2019; Zainuddin and Perera 2019; Zarrinabadi and Ebrahimi 2019).

However, some studies revealed different findings. For example, Strayer (2012) found that students in a flipped classroom were less satisfied with the way that the classroom structure oriented them to learning tasks. Findlay-Thompson and Mombourquette (2014) found no change in the academic outcomes of participants in flipped classrooms. Tse, Choi, and Tang (2019) suggested that flipped classroom instruction had limited capabilities to strengthen the learning motivations of students. A survey of college students showed that students had strong negative views about watching videos before class and the flipped learning environment itself. Through watching videos, learning had some drawbacks, such as students could not immediately ask questions and watching videos was more burdensome than other forms of homework (Xiu et al. 2019). Researchers suggested that teachers could help students overcome their resistance to active learning, such as providing students with abundant resources and adequate guidance to help them prepare for class (Hwang and Lai 2017; Owens et al. 2017). However, this poses a big

challenge for teachers, who find it difficult to give more guidance and timely feedback to students in their spare time. Peer coaching can help solve these problems by providing guidance and feedback to students and helping them to complete extracurricular learning activities.

Peer coaching

As a professional development method, peer coaching has been used in health, business, education and other fields (Arthur 1990; Ruby 2006; Zhang, Liu, and Wang 2017). In the health sector, peer coaching promotes the transfer of knowledge to a clinical environment (Waddell and Dunn 2005) and self-management (Tully et al. 2017). In business, peer coaching was a means for resolving performance problems and for helping employees develop and maintain skills (Arthur 1990). For teachers' professional development, peer coaching involves teachers sharing feedback and giving support to each other for improving their instructional skills and teaching practices (Pearce et al. 2019; Sider 2019). Zhang, Liu, and Wang (2017) summarised four factors of peer coaching for teachers' professional development: academic support, technical support, emotional support and reflective support. Brain and Mirenda (2019) trained nine peer coaches (aged 11–13) for coaching middle school-aged students (aged 12–13) with ASD. The researchers found that peer coaching could strengthen students' engagement, communicative acts, and improve students' social communication skills.

Peer coaching could help learners to address issues through a social interaction with target population who have experiential knowledge and similar characteristics (Dennis 2003), ultimately helping learners achieve success (Arslan and Ilin 2013). Lu (2010) reviewed eight studies of peer coaching applicable in student teacher education and found that peer coaching facilitated learning and improved instructional skills among student teachers, helped student teachers to improve their professionalism, and provided student teachers with affective support. Gribble, Ladyshewsky and Parsons' (2019) research on junior college students (aged approximately 21) showed that peer coaching could improve students' emotional intelligence (EI) and ultimately improve their work performances.

To create successful and sustainable peer coaching, Parker, Kram, and Hall (2014) identified a three-step model of peer coaching, in which step 1 is building the relationship and creating a positive holding environment; step 2 is creating success; and step 3 is internalising the skills. In peer coaching, coaches voluntarily engage in a nonevaluative and mutually beneficial relationship that focuses on the participants' strengths and uses self-evaluation and feedback to build capacity and improve task performance (Stanier 2016).

In this study, peer coaching mainly occurred in two scenarios. The first was when students learned online prior to class. ICT (information and communication technology) and online learning systems were used to support student learning and interaction. The coach tutored students when they were watching videos and observed students' online learning progress and assessed their assignments. Depending on the learning situation, coaches provided feedback to students and helped them understand what they were learning. The second scenario was when students made productions after class. The coach collaborated with the students to complete tasks. When the students encountered difficulties, the coach supported the students and helped them solve problems.

The following research questions were investigated to evaluate the effectiveness of the proposed strategy:

- (1) Can the peer-coach strategy facilitate interaction among group members or among different groups' members in comparison with the conventional video-based flipped classroom?
- (2) Can the peer-coach strategy in a flipped classroom improve student learning achievement in comparison with the conventional video-based flipped classroom?

Method

Participants

The participants were two classes of a college and assigned to the experimental and control group. In the experimental group, there were 44 students, including 10 boys and 34 girls, who were in the 19–24 age range, and the average age was 20.91. There were 43 students including 10 boys and 33 girls in the control group, in the age range of 19–23, and the average age was 20.88. See details in [Table 1](#). In both groups, students were free to form their own small groups. Seven small groups were formed in each class: EG1–EG7 in the experimental group, and CG1–CG7 in the control group. Each group had a leader in charge of the organisation and management of group work. The experimental group learned in a peer-coaching flipped classroom. Students in this group watched microvideos and PowerPoint materials preclass. In class, they were able to propose questions, discuss the materials and practice. Each group was required to make a short film following class. Students in the experimental group were coached by expert students when shooting their videos and editing them for postproduction. The conventional flipped classroom model was applied in the control group. The control group students gained basic knowledge by watching microvideos and PowerPoint materials. These seven groups also made microfilms. The two classes were instructed by the same teacher who had nine years of teaching experience.

Table 1. Information of students of the control group and the experimental group.

	Group	Average age	Members	Male	Female
Control Group	CG1	20.50	T1, T3, T5, T10, T34, T38	2	4
	CG2	21.17	T7, T14, T16, T27, T31, T37	3	3
	CG3	21.50	T2, T8, T9, T40, T42, T43	0	6
	CG4	21.13	T6, T13, T15, T23, T26, T29, T30, T36	5	3
	CG5	20.67	T17, T18, T19, T21, T24, T41	0	6
	CG6	20.00	T4, T11, T22, T25, T39	0	5
	CG7	21.00	T12, T20, T28, T32, T33, T35	0	6
Experimental Group	EG1	20.67	S6, S11, S21, S25, S27, S36	3	3
	EG2	21.00	S2, S7, S8, S15, S19, S26, S41	3	4
	EG3	21.80	S1, S16, S20, S38, S42	1	4
	EG4	20.63	S12, S18, S22, S32, S35, S37, S43, S44	1	7
	EG5	20.83	S4, S17, S23, S30, S31, S34	0	6
	EG6	20.33	S5, S9, S10, S13, S29, S39	1	5
	EG7	21.33	S3, S14, S24, S28, S33, S40	1	5

Instruments

In this study, the pre- and post-test were adopted as measurement instruments. An online cooperative learning environment, the Learning Cell System (LCS) (Yu et al. 2015), was used to support the implementation of online learning activities and interactions. The LCS¹ could support online interaction and discussion, collaborative knowledge editing, and activity monitoring (Yang, Guo, and Yu 2016). In this study, the teacher uploaded the instructional videos and other materials to the LCS and arranged learning activities. Students were expected to watch the videos and use other materials, along with participation in the learning activities, to learn the concepts and complete tasks at their own pace out of class.

Before the experiment, the pre-test was administered to evaluate the students’ prior knowledge of the course. After the experiment, the post-test was undertaken to evaluate the students’ academic performances. The pre-test and post-test consisted of 20 one-choice items (20%), 15 multiple-choice items (30%), 10 fill-in-the-blank items (20%), and 5 question-and-answer items (30%). A perfect score on the pre-test and post-test was 100.

Procedure

The experiment lasted six weeks, and the procedure is shown in Figure 1. Both experimental and control group students took the pre-test in the first week. Students in both groups used the same learning materials out of class, and materials were provided by the same teacher. Learning materials mainly include microcourse videos, teaching PPT and other reference materials. The teacher recorded approximately 15 minutes of teaching videos (see Figure 2), mainly introducing the knowledge of the course. The teacher uploaded videos and other learning materials to the LCS. Students watched videos prior to class to gain knowledge. In addition, the teacher provided teaching PPT materials and extracurricular reference materials for learners. Figure 3 shows the teaching PPT

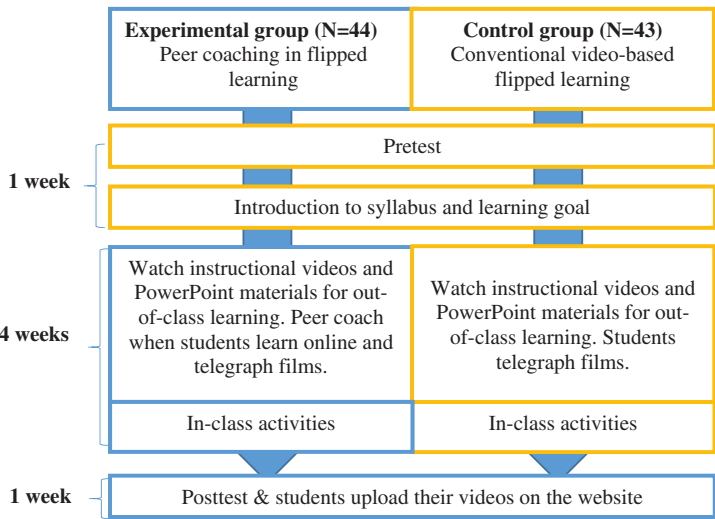


Figure 1. Diagram of the experimental design.



Figure 2. Instructional video for online learning.

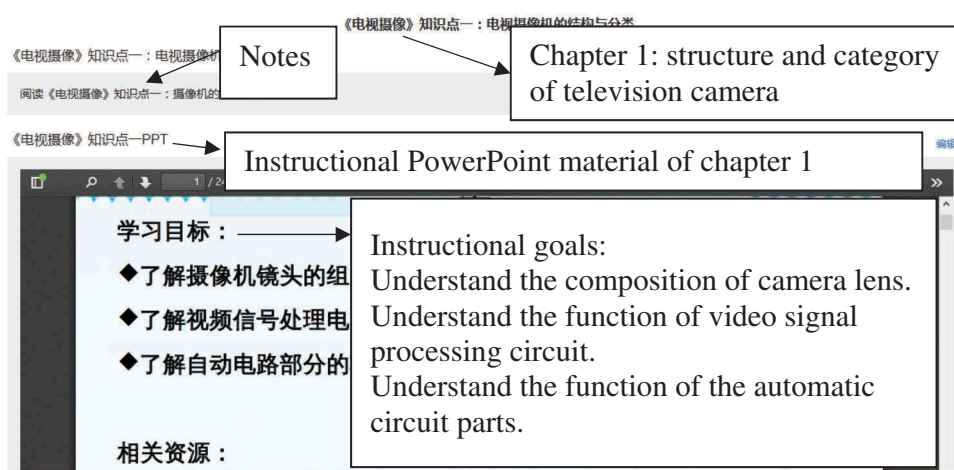


Figure 3. Instructional PowerPoint material for online learning.

materials provided by the teacher. In the first class, the teacher introduced the syllabus and learning goal, i.e. that students should construct a microfilm script and make a microfilm (about ten minutes) in small groups. Moreover, the teacher recruited seven outstanding students as tutors to coach the seven subgroups in the experimental group. These seven tutors (S7, S22, S28, S30, S36, S39, and S42) were also team leaders and received training regarding professional knowledge and skills by the teacher. The teacher asked all students to help each other when they had problems while learning. In the control group, the teacher did not recruit students to coach other students. Students in this group could communicate and interact freely and ask each other for help in or between groups.

In the experimental group, the tutors mainly provided online learning assistance when students were learning about the video materials outside of the classroom. When students had problems, they could consult with tutors. The tutors discussed and explained learning content with the students. The tutors would also provide learning materials to help students understand the concepts and gain knowledge. In addition, students

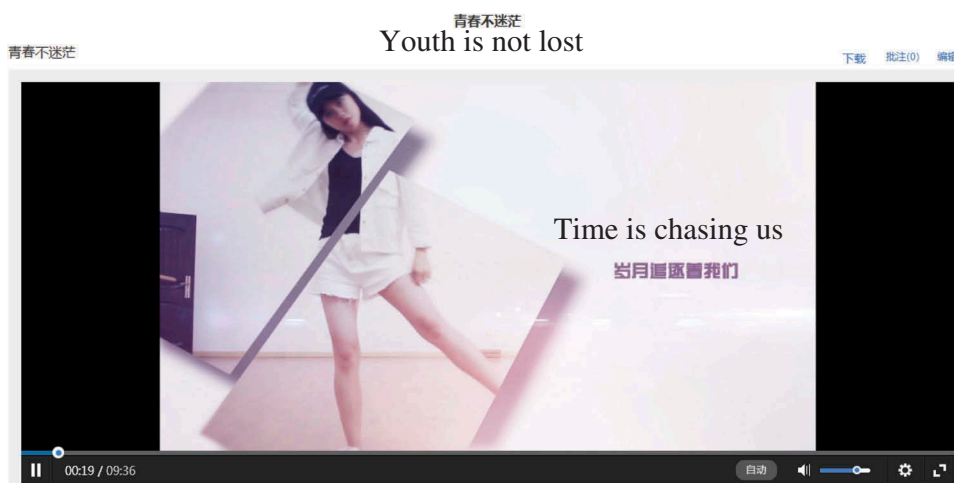


Figure 4. Video works of one collaborative group.

interacted with other students besides tutors. Out of class, the tutors discussed script-writing and video filming with group peers. During filming, the tutors observed students' behaviours and provided help when students made errors or did not know how to operate the camera. The tutors explained the steps to the students or instructed students about how to use the camera and nonlinear editing software. In class, the teacher taught the class, facilitated student discussions and helped students solve problems. The tutors studied with group peers, participated in class discussions and answered questions raised by students and teachers. In the sixth week, both the experimental and control groups took the post-test and uploaded their works to the LCS. [Figure 4](#) is an example work of one group. The film name is *Youth is Not Lost*.

Data coding and analysis

The study mainly used a social network analysis method to analyse the learning interactions. There were two data sources for learning interactions. The LCS enables students to communicate and discuss online. Students posted and replied to posts in the forum of the LCS. For example, student S1 posted a post and student S16 replied to S1, an interactive relationship was established between S1 and S16. The interaction frequency of S1 and S16 was 1. The number of interactions is used to indicate the strength of the interactive relationship. In the after-class practice (shooting works), students had offline interactive behaviours. We used surveys to document offline interactions between students. On the weekend of every week, we asked students to write down with whom and how many times they interacted during the week. Data were collected four times over four weeks, and the data of the four weeks were added according to the corresponding students. Finally, the total interaction frequency of students was obtained by adding the online interaction frequency and offline interaction frequency.

Quantitative analysis was used in this study. Centrality (including degree, closeness and betweenness) of individuals and groups was calculated. Centrality refers to a group of

metrics that aim to quantify the ‘importance’ or ‘influence’ of a particular node (people or group) within a network (Tsvetovat and Kouznetsov 2011). The measurements of node centrality in common use are degree centrality, betweenness centrality, and closeness centrality (Opsahl, Agneessens, and Skvoretz 2010). Degree centrality is a simple count of the total number of connections linked to a vertex. Closeness centrality is usually interpreted either as a measure of access efficiency or of independence from potential control by intermediaries (Brandesa, Borgattib, and Freeman 2016). Betweenness centrality measures the number of times that an actor is on the shortest path between other pairs of actors (Freeman, White, and Romney 1992), and it represents the control of one actor over other pairs of actors; the ability of some actors to control (or ‘broker’) the flow of connectivity (information, resources, etc.) within the network, or to connect otherwise disconnected groups (Hughes, Bright, and Chalmers 2017).

The t-test and chi-square test were carried out for the experimental group and the control group, respectively, from the individual level and group level to understand the interaction differences between the two groups. To understand the network structure, the social network diagrams of the two groups were drawn. Finally, the one-way analysis of covariance (ANCOVA) was carried out on the learning performance of the experimental group and the control group, and ANOVA was conducted on the learning performance from the two levels of closeness centrality and betweenness centrality, to explore the influence of interaction on learning performance.

Results

UCINET, Gephi and IBM SPSS were employed to analyse the data, and examine the performance of the two groups, including their pre- and post-tests, centralities and learning performances.

Analysis of learning interaction

Interaction between student individuals

Table 2 shows the t-test results of the centralities of the two groups. For degree centrality, the means and standard deviations were 15.64 and 5.12, respectively, for the experimental group, and 10.85 and 3.13, respectively, for the control group students. The t-test results indicated that the degree centralities of the two groups were significantly different ($t = 5.28$, $p < 0.001$, Cohen’s $d = 1.14$). The value of effect size (d) shows a large effect (Cohen 1988).

Table 2. The t-test of centralities for the two groups.

Variable	Group	N	Mean	SD	t	d
Degree	Experimental group	44	15.64	5.12	5.28***	1.14
	Control group	43	10.85	3.13		
Closeness	Experimental group	44	10.72	3.30	15.85***	3.41
	Control group	43	2.73	0.50		
Betweenness	Experimental group	44	2.75	6.02	2.81**	0.60
	Control group	43	0.19	0.65		

** $p < .01$, *** $p < .001$.

The experimental group obtained a significantly higher value for the closeness centrality ($M = 10.72$, $SD = 3.30$) than the control group ($M = 2.73$, $SD = 0.50$). The t-test indicated that the closeness centrality of the two groups was significantly different ($t = 15.85$, $p < 0.001$, Cohen's $d = 3.41$). Furthermore, the effect size (d) was large. These results indicated that students in the experimental group had more paths to get information and knowledge from other students than the control group.

For the betweenness centrality, a t-test showed that the difference between the traditional flipped classroom and the peer-coaching flipped groups was significant ($t = 2.81$, $p < 0.01$, $d = 0.60$). The experimental group students ($M = 2.75$, $SD = 6.02$) performed better than the control group ($M = 0.19$, $SD = 0.65$). There were more experimental group students playing a role as intermediaries, and more students had control ability compared to control group students.

To get an intuitive understanding of the collaborative interaction between students of the two different groups, the social network structure graphs of the groups were drawn by Gephi software. Node and edge size were set by degree centrality (see Figures 5 and 6). There were only two teams (CG4 and CG6) that had interactive relations in the control group. However, only one student (T22) served as a mediator connecting these teams. There was a low robustness of the network structure for CG4 and CG6 teams, and the structure was unstable. The other five teams were isolated, and students interacted only

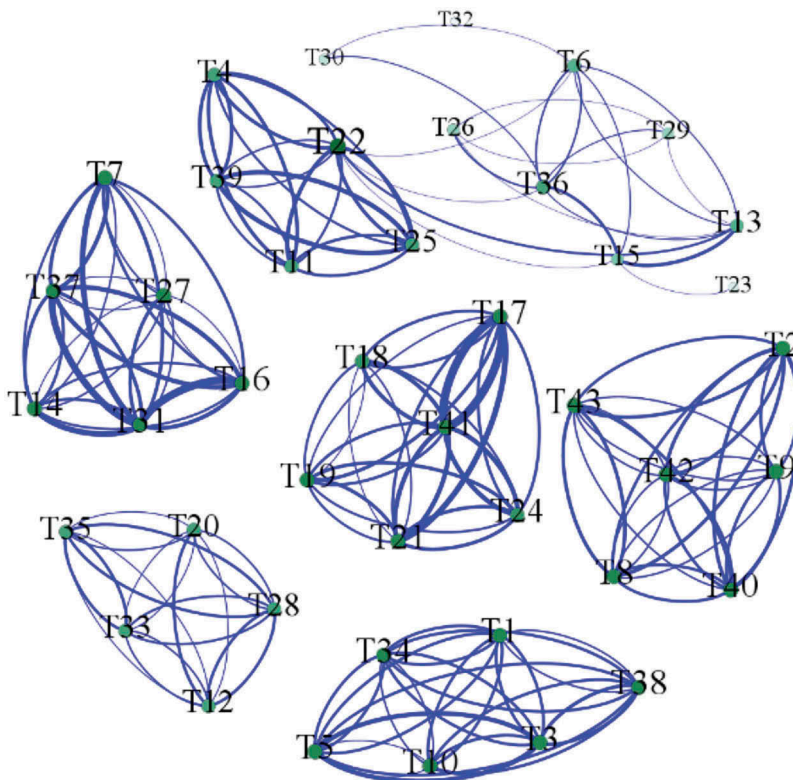


Figure 5. Social network of control group students.

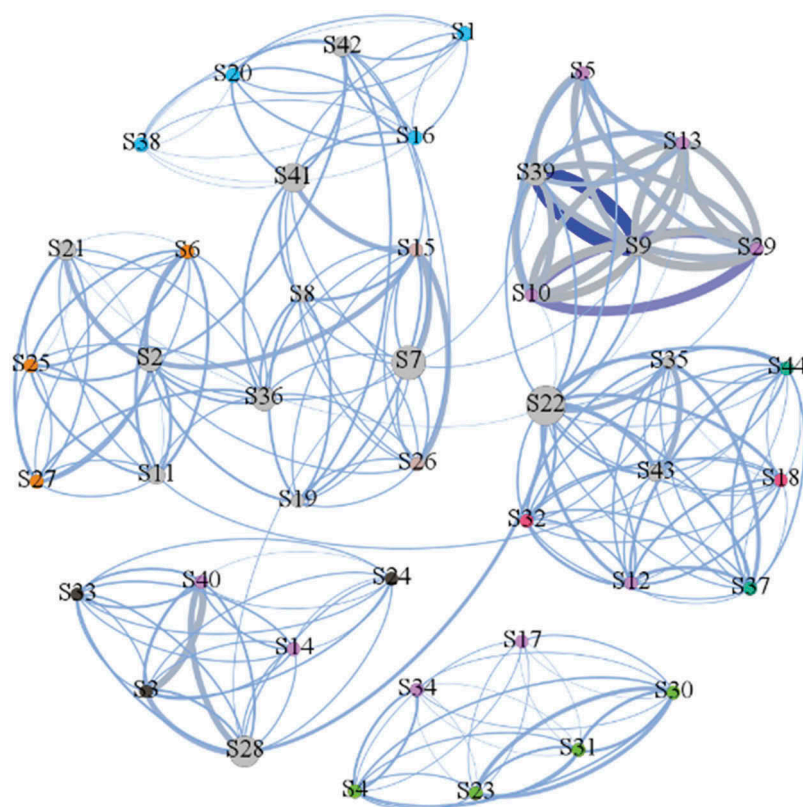


Figure 6. Social network of experimental group students.

with team peers. Every team was closely coordinated, which means that interactions were mainly between team peers in the control group.

In the experimental group, students such as S22, S7, S36, S41, and S42 had larger nodes than the other students. These students played mediation roles for enhancing team interaction. They were bridges of team communication, and important structural openings of the whole network of the experimental group. For example, tutor S7 interacted with students who came from EG1, EG2, EG3, EG6 and EG7, and connected these teams. Tutor S22 connected the teams EG1, EG4, EG6, and EG7. In [Figure 6](#), tutors S7, S22, S28, S36, S39, and S42 had more interfaces with students who came from the other teams. Tutor S30 only interacted with his team's peers. However, some students (e.g. S2, S29, and S41) who were not tutors also had more links with students who were not team peers. These students enhanced dissemination of information and knowledge. They could offer help and instruction to other students. Students interacted with these class members by consulting them and asking for help with problems. In the experimental group, tutors interacted not only with team peers but also with other student groups. The tutors connected the small groups and enhanced interactions between them.

Interactions between student teams

[Table 3](#) shows the results of centralities for teams in the two classes. In the control group, only two teams, CG4 and CG6, had degree and closeness values. For all teams, betweenness

Table 3. Centralities of student teams of the two groups.

	CG1	CG2	CG3	CG4	CG5	CG6	CG7
Control group							
Degree	0	0	0	16.67	0	16.67	0
Closeness	-	-	-	16.67	-	16.67	-
Betweenness	0	0	0	0	0	0	0
	EG1	EG2	EG3	EG4	EG5	EG6	EG7
Experimental group							
Degree	33.33	66.67	16.67	50	0	33.33	33.33
Closeness	40	46.15	35.29	40	-	40	40
Betweenness	4.44	36.67	0	10	0	4.44	4.44

values were zero. In the experimental group, EG2 had the highest degree, closeness and betweenness centrality. This means that EG2 was the centre node of the group network. Betweenness of EG3 was zero, which means that EG3 did not connect two or more teams. Degree and betweenness values of EG5 were zero, and its closeness value was null, indicating that it was an isolated node.

The chi-square test was employed to examine the differences between the control and experimental groups (see Table 4). For degree centrality, there was a significant difference between the two groups ($\chi^2 = 4.667$, $p < 0.05$), indicating that teams of the experimental group had more interactions than the control group teams. For closeness, there was also a significant difference ($\chi^2 = 4.667$, $p < 0.05$), which means more teams had a short distance from each other in the experimental group. The experimental and control groups also showed a significant difference ($\chi^2 = 7.778$, $p < 0.01$) for betweenness centrality, as there were more mediators for teams in the experimental than control group. In the experimental group, more intergroup interactions were implemented between teams, and the frequency of interaction was higher than for the control group. In the control group, members of the team interacted with each other to complete the task together, but students had less interaction among teams.

Gephi software was used to analyse the teams' interaction data and draw the social network structure of the groups (see Figure 7). The property of group nodes and edges were set according to the degree centrality. In Figure 7, the group EG2 connected with six teams. The edges between EG2 and EG1 and the edges between EG2 and EG3 were bolder than other edges, indicating that EG2 had more frequent interactions with EG1 and EG3. Information and knowledge were transmitted frequently between EG1 and EG2, EG2 and EG3. EG5 was an isolated node. This is consistent with the results of the previous

Table 4. The chi-square test of centralities for the two groups.

Centrality		Value	df	Asymp. Sig. (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)
Degree	Pearson Chi-square	4.667 ^a	1	.031		
	Fisher's Exact Test				.103	.051
Closeness	Pearson Chi-square	4.667 ^b	1	.031		
	Fisher's Exact Test				.103	.051
Betweenness	Pearson Chi-square	7.778 ^c	1	.005		
	Fisher's Exact Test				.021	.010

^a4 cells (100%) have expected counts less than 5. The minimum expected count is 3.00.

^b4 cells (100%) have expected counts less than 5. The minimum expected count is 3.00.

^c4 cells (100%) have expected counts less than 5. The minimum expected count is 2.50.

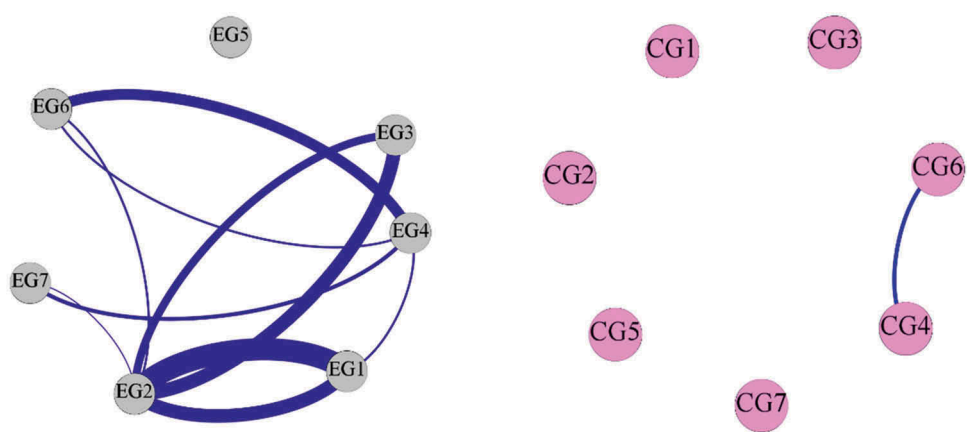


Figure 7. Social network structure of groups for the experimental group (left) and the control group (right).

study of degree and betweenness centrality values. There were five teams functioning as isolated nodes in the control group. Only CG4 interacted with CG6. However, there was only one edge between them, which indicated that information and knowledge was transmitted between the two teams.

Analysis of learning achievement

To evaluate the effectiveness of the peer-coaching flipped classroom, students' learning achievement in the television shooting course was compared. The t-test results from the pre-test showed that for the experimental group, the mean value was 87.32, and the standard deviation was 4.74, while for the control group, the mean value was 87.86 and the standard deviation was 5.02. There was no significant difference between the two groups ($t = -0.518$, $p > 0.05$). The t-test results of the pre-test indicated that the two groups had equivalent prior knowledge.

To avoid any influences between the pre-test and post-test, the ANCOVA was employed to evaluate students' learning achievement in the experimental group and in the control group by the pre-test scores as the covariate and the post-test scores as the dependent variables. The Levene's test of determining homogeneity of variance was not violated ($F = 1.19$, $p > 0.05$), indicating that the assumption is tenable and the ANCOVA results can be used to interpret the relationships between the students' tested prior knowledge and their learning achievement in the post-test. Table 5 shows the results of the learning achievement according to the post-tests of the two groups. The adjusted means and standard deviation of the post-test scores were 81.18 and 1.51, respectively,

Table 5. Results of one-way ANCOVA for post-test scores.

Group	N	Mean	SD	Adjusted mean	Adjusted SD	F
Experimental group	44	81.00	9.32	81.18	1.51	16.05***
Control group	43	72.77	11.43	72.59	1.52	

*** $p < 0.001$.

Table 6. The correlation of the students' centralities and learning achievements.

	Degree	Closeness	Betweenness	Post-Test
Degree	1			
Closeness	0.609**	1		
Betweenness	0.699**	0.378**	1	
Post-Test	0.180	0.395**	0.229*	1

* $p < 0.05$, ** $p < 0.01$.

for the experimental group, and 72.59 and 1.52, respectively, for the control group. The results ($F(1,85) = 16.05$, $p < 0.001$) showed a significant difference between the two groups. The experimental group's post-test scores were significantly higher than those of the control group. This implies that the students who learned in the peer-coaching flipped classroom showed significantly better learning performance than those who learned in the conventional flipped classroom.

To further understand the interaction on the learning achievement of the students, Pearson's correlation coefficient analysis was employed to evaluate the correlation of degree, closeness, betweenness and learning achievement. Table 6 shows clearly that learning achievement had a positive correlation with closeness and betweenness, but a weak correlation with degree.

Because learning achievement correlated with closeness and betweenness, this study further explored the effects of interaction on the learning achievement of students with different closeness and betweenness levels. The study evaluated the interaction between the different learning approaches (peer-coaching flipped classroom and conventional flipped classroom) and different closeness and betweenness levels (higher and lower) by analysis of two-way ANOVA. The assumption of homogeneity of regression was not violated for closeness ($F = 0.42$, $P > 0.05$) and betweenness ($F = 0.27$, $P > 0.05$), suggesting that a common regression coefficient was appropriate for the two-way ANOVA (see Tables 7 and 8). The results showed that learning approaches ($F = 13.74$, $P < 0.05$) significantly affected learning achievement. However, the closeness ($F = 0.02$, $P > 0.05$) and the interaction between the learning approaches and closeness ($F = 3.98$, $p > 0.05$) had no significant effects on learning achievement. Table 8 showed that betweenness and interaction between betweenness ($F = 0.28$, $p > 0.05$) and learning approach ($F = 1.69$, $p > 0.05$) had no significant effects on the learning achievement.

Table 7. The two-way ANOVA result of the learning achievement and closeness.

Variable	SS	df	MS	F	P	η^2
Learning approach	1456.56	1	1456.56	13.74	0.00	0.14
Closeness group	2.25	1	2.25	0.02	.89	0.00
Learning approach * Closeness group	422.03	1	422.03	3.98	.05	0.05
Error	8798.62	83	106.01			

Table 8. The two-way ANOVA result of the learning achievement and betweenness.

Variable	SS	df	MS	F	P	η^2
Learning approach	1456.56	1	1456.56	13.42	0.00	0.14
Betweenness group	30.20	1	30.20	0.28	0.60	0.00
Learning approach * Betweenness group	183.58	1	183.58	1.69	0.20	0.02
Error	9011.62	83	108.57			

In this study, the social network centrality of learners was related to the interaction frequency and interaction structure of learners but had little correlation with the interactive content. However, test scores indicated that learning performance was related to interactive content. In the television shooting course, student interaction in the classroom was related to learning content. Students collaborated to film and edit videos, communicating with each other about how to use the cameras and edit the videos with Adobe Premiere software and Adobe After Affect software. However, for most of the time, student interaction was more meaningless as they worked on their microfilms.

Discussion and conclusions

This study explored how a peer-coaching approach could be used to improve student engagement and interaction to help them gain knowledge in a flipped classroom. In the peer-coaching flipped classroom, some outstanding students coached other students in a television shooting course. Students were encouraged to ask for help and give help to each other to make a high-quality film. This study showed that a peer-coach strategy could promote students' engagement and interaction to achieve better learning achievement than in traditional flipped classroom approaches.

In this study, peer coaching in a flipped classroom was found to promote the students' interaction. This finding was noted not only in increased interactions among students within the collaborative group but also in the interactions among the teams. The findings showed that peer coaching could encourage the tutees to participate in the discussion and to pose questions to each other to develop their understandings of the learning content (Quoß, Rüttermann, and Gerhardt-Szep 2017). Peer coaching could improve interpersonal skills of both tutees and tutors, enhance self-worth, increase motivation for learning, and promote greater active learner engagement (Altintas, Gunes, and Sayan 2016). However, the study showed that the closeness and betweenness had no significant impacts on academic performance. The interactive content and quality are related to academic performance. The centrality only indicates the structural position of students, but the quality of the interaction is not involved. This study only discussed the relationship between centrality and academic performance but did not analyse the relationship between the content of the interaction and the academic performance, which was also a limitation of the study.

Based on the analysis of individual and team interactions, peer coaching led to a higher frequency of interaction in the flipped classroom. However, there were not many lines between nodes, especially as there were not many two-way edges between nodes. This indicated that students could not negotiate with each other for many rounds. Most students only got assistance and information from tutors and other students.

The learning achievement results showed that peer coaching improved student learning. Professional tutors could play a role as teachers, provide effective instruction to students, and provide feedback to tutees in a timely way. In the practice of telegraphy and postproduction editing, tutors provided technical assistance and help to students to solve problems, ultimately improving students' professional abilities with photography and postproduction. This result had been confirmed in other studies. For example, Sider (2019) pointed out peer coaching supported professional capacity development. Hohensee and Lewis (2019) stated that peer coaching could build

conceptual disciplinary knowledge. Peer coaching could also improve medicine students' professional abilities (Fried, Atkins, and Irwin 2019) and help them develop from novices to advanced beginners with higher-level clinical experiences (Menard and Maas 2019).

In summary, this study suggests that the peer-coaching flipped learning approach is more instructionally efficient than a traditional flipped classroom approach for the television shooting course. The major contribution of this study is to provide evidence that integrating the peer-coach strategy into flipped classrooms can promote students' interaction and knowledge construction, thus further improving their learning achievements. Limitations of the current study should also be noted. The experiment duration was short (6 weeks), and the course was carefully structured. Instruction in other contexts may not be as amenable to the peer-coaching flipped learning approach. In addition, information about students' offline interactions was lacking, owing to the limitation of the data collection tools and measurement tools. This may have led to some bias in the results. Future research could record students' offline interactions as analysis of this aspect may improve the validity of the research.

Note

1. <http://www.etc.edu.cn>.

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